Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with **3 years** of industry experience building and operating **production RAG and tool-calling agent systems** at an asset manager (**HDFC Asset Management**), where agents called Python functions to run fund-comparison calculations for a mutual fund house, behaviour was guardrailed through **system prompts versioned in Git** and every release was reviewed by **Compliance**. Weighed **cost against quality** between self-hosted **Llama-3** and **Azure OpenAI**, and ran an MSc **benchmark study** of deep learning architectures (five seeds, pre-trained VLMs as baselines). **MSc in Data Science** at the University of Bristol (awaiting result), with an **AstraZeneca** dissertation whose ranking was refined on scientist feedback. Seeking an **AI Engineer** role building and evaluating **RAG and agentic systems**.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

RAG & Agentic Systems in Production

- Designed and built production **RAG** applications end to end (**ReactJS** frontend, **Python FastAPI** microservices, retrieval with **LlamaIndex** and **Azure OpenAI GPT-3.5 & Llama-3**) for Compliance, Client Service and Investment, with a separate **pgvector** database and **Azure OpenAI** embeddings per department so access stayed siloed and similar content from different departments was not confused.
- Built **agentic tool-calling** where AI agents call **Python functions** as tools to run SQL-backed fund-comparison calculations and generate PDF comparison reports on request, on top of a **PostgreSQL** platform queried with **Python Pandas**.
- Took applications from prototype to real users and iterated on what broke; guardrailed behaviour through **system prompts** (no fine-tuning), **versioned every prompt change in Git** so changes stayed traceable, and had **Compliance** review each release.
- Made the **cost-versus-quality call** on model tooling: self-hosted **Llama-3** on a **GPU EC2** instance for the largest department (Client Service) and used stronger **Azure OpenAI** outputs for decision makers, keeping the ability to switch models.

MLOps & Deployment

- Designed the **AWS** infrastructure for internal analytics platforms and GenAI applications (**EC2**, **S3**, **ECR** and **Lambda**), packaged the FastAPI microservices as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab** CI/CD pipeline.
- Implemented **RBAC** with **LDAP** authentication, restricted access to company private IPs with **AWS Security Groups** and deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

Data Pipelines, Analytics & Stakeholder Delivery

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.

- Built a **Python** pipeline that parses over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, **saving 60 analyst hours a month**; scheduled on **AWS SageMaker** with EventBridge and Lambda, and replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images.
- Analyzed customer behaviour using **Python Pandas** and **Market Basket Analysis (Apriori Algorithm)** to surface buying patterns for the marketing team; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** and alongside engineers and interns to turn requirements into working tools; **de facto team lead for up to 4 interns**; **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: the **scoring and ranking algorithm** that ranks cancer cell lines by target inclusion and exclusion genes using per-gene scores across six evidence layers, calibrated across lineages and grouped into confidence tiers; the **data preprocessing pipeline** that builds 14 clean, joined tables from 17 raw inputs with integrity checks; and the **RAG-based AI assistant**. Ranking validated with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. **Multi-architecture deep learning study** comparing CNN, ResNet and ViT encoders trained from scratch with symbolic, SBERT-retrieval and LSTM/attention decoders for generating image descriptions on three synthetic datasets, **evaluated across five seeds** with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs, noting that the protocols differ so the comparison is not like-for-like.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Wingman:** AI dating-coach web app with **108 LLM personas** across 12 archetypes and a vision model that scores uploaded photos (ReactJS, Python FastAPI, NVIDIA NIM). LLM output treated as untrusted: **validated with Pydantic, repaired and cached by input hash**; cost and rate limits shaped the design (12 LLM calls instead of 108, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolutional Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2 incl. GPU, S3, ECR, Lambda, SageMaker, EventBridge), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite